

SUSHANT LOKHANDE

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EXPERIENCE

San Jose State University | Graduate Research Assistant, Retrieval & Ranking Aug 2025 to May 2026

- Built an automated evaluation harness for controlled experiments, cutting setup effort **55%** and making every result reproducible from a single config.
- Boosted malware classification to **80.2%** across 17 families with a soft-voting **PyTorch** ensemble (CNN, ResNet50, ViT), **2.4 pts** over the best single model at $p < 0.001$.
- Reduced rebuffering **19%** versus a fixed-rate baseline by building a **Docker**, **WebTransport**, and **Media-over-QUIC** testbed and running controlled adaptation experiments across network and protocol configs.
- Mentored **200+ students** across **3** courses (Programming Paradigms, AI, ML) through debugging sessions, code reviews, and grading.

Turito | Software Engineering Intern, Full-Stack Product Jun 2022 to Oct 2022

- Cut controlled-disruption failures **42%** (3.1% to 1.8%) by hardening **retries**, timeouts, failure recovery, and distributed state transitions.
- Slashed p95 **CPU 22%** by profiling high-frequency media paths and eliminating excess buffer allocation, payload copies, repeated routing lookups, and **lock contention**.
- Reduced production diagnosis time **35%** by adding structured **metrics**, runtime-state visibility, and targeted logs.
- Kept live classes running for students and teachers across about **12 regions** by cutting session failures **42%** (3.1% to 1.8%) on a **WebRTC** classroom platform.

Trivia Softwares | Software Engineering Intern Nov 2021 to Jan 2022

- Built a **Rust/Tokio** load-testing framework driving about **25,000** concurrent sessions, exposing lock contention, serialized **WebSocket** writes, and slow session cleanup.
- Cut performance-investigation time about **40%** by building tooling on Rust workers, **Python** aggregation, **SQLite**, Flask, and Chart.js.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, C++17, SQL

Search & Ranking: HNSW vector search, two-tower retrieval, ANN candidate generation, neural ranking, Recall/NDCG/MAP evaluation

Full Stack: React, Next.js, FastAPI, REST, gRPC, Redis, PostgreSQL, MongoDB

Infra: Docker, AWS, Kafka, pytest, Locust, Selenium, CI/CD

PROJECTS

ReelRank, Two-Stage Recommender | *PyTorch, C++ HNSW, FastAPI, React, TypeScript*

- Built a **two-tower** retrieval model with ANN candidate generation and a neural ranker, evaluated on a leakage-free temporal split with Recall, NDCG, and MAP.

Proxima, Vector Search Engine | *C++17, AVX2/SIMD, HNSW, pybind11*

- Reached **0.999 recall** at **12,049 QPS** single-threaded, **1.8x** hnswlib and **2.5x** FAISS, with an HNSW engine written from scratch in C++17.

Relay, LLM Inference Gateway | *Python, FastAPI, asyncio, SQLite, SSE*

- Cut p50 latency from **759 ms to 44 ms** over a **4,000**-request load test with an OpenAI-compatible gateway doing exact and **semantic caching**, failover, and rate limiting.

OPEN SOURCE

- microsoft/markitdown**: fixed EPUB conversion for percent-encoded manifest paths, with regression tests (merged)
- fission/fission**: 3 merged PRs, Kubernetes event and KEDA object capture in support dumps, removed duplicate package enumeration

EDUCATION

San Jose State University | M.S. in Computer Science | GPA 3.8/4.0 | Spartan Scholar Aug 2024 to May 2026

Coursework: Database Systems, Advanced Algorithms, Machine Learning, Distributed Computing, NoSQL, Computer Networks

University of Mumbai | B.Tech. in Computer Engineering | GPA 3.9/4.0 Jun 2019 to May 2023

Research: malware image-classification ensemble selected for Dr. Mark Stamp's ML textbook; sole-author contactless palmprint verification (0.9975 AUC)